

Exploring Generic Global Happiness through Human Motivation and Maturity of Heart: An Explainable AI Analysis

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Abstract. The psychological needs, as well as moral-emotional development, influence cross-national differences in happiness; however, the comparative role of these two dimensions has not been examined in the cross-cultural context. This study presents a systematic Maturity of Heart model, including ethical reasoning, empathy and compassion, civic engagement, relational trust, and personal growth, and then combines these aspects with the Maslow's needs to predict the subjective well-being. Survey variables were operationalised as theoretical constructs, processed and analysed through descriptive and correlational analysis using the World Values Survey Wave 7 dataset, which includes about 97,000 respondents in 65 different countries. Variance Inflation Factor analysis and recursive feature elimination using cross-validation was used to select the most informative predictors. Several predictive models including the ensemble machine learning algorithms and deep learning networks were evaluated; CatBoost returned the best overall predictive accuracy (87.78%). An explainable AI analysis through SHAP indicated which core needs (self-actualisation, physiological needs and safety, in particular) have the most significant impact on happiness, and Maturity of Heart dimensions (community trust and ethical reasoning in particular) have complementary effects. Those findings support the Maturity of Heart framework, demonstrate the usefulness of theory-driven predictive modelling, and provide more practical implications of cross-cultural well-being research and policy interventions.

Keywords. Happiness; Subjective well-being; Maslow's needs; Maturity of Heart; Explainable AI; Machine learning; Deep learning; CatBoost.

INTRODUCTION AND BACKGROUND

Happiness has been recognised as a multidimensional construct that goes beyond economic measures such as gross domestic product and includes subjective well-being (SWB) and more general notions of human flourishing. The modern research highlights that, happiness not only includes material states, but also psychological satisfaction, socialisation and values-derived meaning, especially when it is studied within the perspective of the different cultures [1], [2]. The full interpretation of global happiness hence requires theoretical models that integrate the primary human requirements as well as the level of advanced psychological and moral development. One of the most influential frameworks in well-being research, Maslow's theory of human motivation, conceptualises happiness as being associated with fulfilling multiple human needs, including needs in the physiological, safety, belonging, esteem and self-actualisation domains [3]. While being widely applied in empirical studies, Maslow's framework is often interpreted hierarchically and might fail to adequately reflect the non-linear and culturally contingent nature of well-being, especially about moral-emotional and relational dimensions that are at the heart of long-term eudaimonic happiness [4], [5], [6].

In order to overcome these limitations, this research combines Maslow's generic needs with a Maturity of Heart model, and operationalises psychological, moral, and social potentials to enable sustainable happiness. The five dimensions of the framework are ethical and moral reasoning, empathy and compassion, civic engagement and social responsibility, relational maturity and trust, and personal growth [5], [7]. In so doing, it places maturity in a mediating role between motivation and long-term well-being and explains why people with similar needs or intrinsic drives can have different degrees of happiness depending on their moral, emotional, and social development [8], [9]. This method is consistent with Self-Determination Theory (SDT) which argues the relevance of intrinsic motivation and value-congruent action in promoting eudaimonic well-being, and generalises it by including the relational and ethical aspects often lacking in motivation-only analyses [10], [11].

Empirical research on global happiness typically employs massive cross-cultural data, especially the World Values Survey (WVS), which measures life satisfaction and affective well-being in many countries [12], [1], [13]. Measures of life satisfaction operationalise the evaluative components of happiness, whereas affective components indicate daily emotional experiences, which capture both hedonic and eudaimonic elements [14], [10]. Research based on WVS data shows happiness is not limited to material and motivational factors but also to social trust, support from relationships and civic participation, which points out the dynamics between an individual's values and society [15], [16], [17]. These findings highlight the shortcomings of motivation-oriented models, where intrinsic motivation factors alone could not explain cross-cultural differences in the outcomes of happiness [18], [14].

Maturity acts as an important moderator in this process. The psychological and emotional maturity, which is reflected in self-reflection, emotional control and resilience, is supportive of the stable well-being even in less favourable conditions [9], [19], [8]. Ethical and moral maturity allows individuals to connect actions to their internalised principles which leads to prosocial behaviour, relational trust and long-term life satisfaction [5], [20]. Social and civic maturity involves taking these capacities beyond the self, connecting the inner development of the individual to the community to which he or she belongs and the collective well-being of that community, and showing that mature people translate internal motivation into socially meaningful outcomes [7], [21]. Collectively, these domains offer a coherent view on the developmental influence of maturity on happiness, overcoming shortcomings of previous studies that examined psychological, moral and civic dimensions separately [22].

Advances in analytical methods have allowed greater subtlety in the modelling of these complex relationships. Machine learning techniques, such as ensemble methods, are able to capture the non-linear interactions between the motivational, maturational and socio-cultural variables that go beyond what is possible with traditional regression methods for predicting life satisfaction [23], [24]. Deep learning further detects latent features in high dimensional data, but is limited in the degree of interpretability [25], [26]. Explainable artificial intelligence (XAI) methods, such as SHapley Additive exPlanations (SHAP), fill this void by uncovering the relative contributions of psychological, social and cultural predictors [27], [28]. This approach connects the outputs of the models with theory and helps ensure that findings are interpretable and can be applied across diverse populations [15].

Combining Maslow's generic needs and the Maturity of Heart framework, as well as explainable machine learning, this research offers a unified approach to defining global happiness. It aims at determining not only what motivational and maturational factors lead to well-being but also the interaction between these factors to bring about a longer-lasting, meaningful, and culturally based happiness. In doing so, the research fills in theoretical gaps, draws on strong empirical evidence and employs advanced analytical methods to examine the association of human motivation, maturity and social context in a single interpretable model of global subjective well-being [5], [10], [14].

DATA AND METHODOLOGY

This research uses both theory-based and data-based research approaches to examine the connection between human motivation, maturity of heart, and cross-national happiness. The methodological framework combines a vast amount of survey data with machine-learning (ML), deep-learning (DL), and explainable-AI (XAI) procedures to evaluate empirically the developed theoretical model. Happiness is operationalised as subjective well-being (SWB), consistent with established literature that treats self-reported life satisfaction as the central indicator of overall happiness.

Data Source and Variable Operationalisation

The present analysis uses data from Wave 7 of the World Values Survey (WVS, 2017 - 2022), which includes approximately 97,000 respondents in 65 countries [29]. The WVS provides in depth information on socio-demographic variables, values, beliefs and subjective well-being and therefore enables cross-cultural investigations of determinants of happiness.

Survey variables were mapped in line with theoretical constructs based on Maslow's needs and Maturity of Heart framework. Maslow's needs were operationalised in five domains: physiological needs, safety needs, love and belonging needs, esteem needs, and self-actualisation needs. The framework of Maturity of Heart includes five dimensions, namely ethical or moral reasoning, empathy or compassion, civic engagement or social responsibility, relational maturity or community trust, and personal growth, which capture the moral-emotional and social aspects of human development (**Figure 1**).

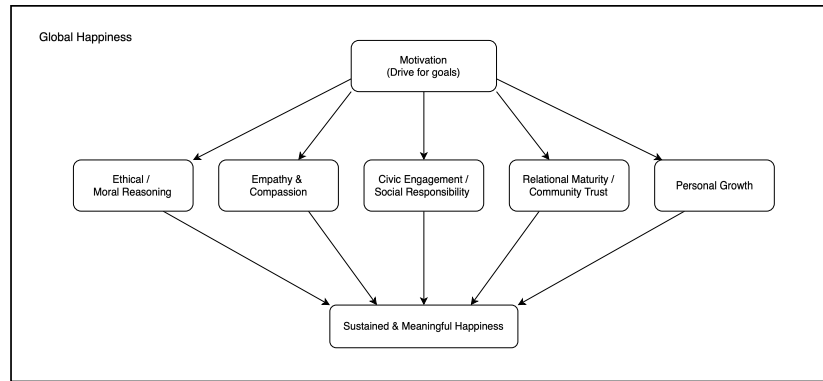


FIGURE 1. *Maturity of Heart Conceptual Framework (illustrating the five dimensions through which internal values, ethical awareness, and social behaviour mediate the relationship between motivation and global happiness).*

The five dimensions of the Maturity of Heart framework capture complementary aspects of moral-emotional and social development. Ethical and moral reasoning refers to the ability to evaluate actions in terms of fairness, integrity, and justice which is not restricted to self-interest. Compassion and empathy denote awareness of the needs and suffering of others, especially the disadvantaged groups hence resulting in social cohesion. The concept of civic engagement and social responsibility implies the active involvement in the community and societal processes such as trust in institutions and the collective welfare. Relational maturity and community trust is the capacity to create and sustain successful interpersonal relationships and social bonds. Lastly, personal development entails personal reflection, autonomy, and matching personal values with behaviour thus sustaining a life-long psychological growth.

To ensure the quality of the data, variables were cleaned, recoded and scaled. Missing values were treated by use of simulation-based imputation strategy and the comparison of mean and median imputation. Ordinal, nominal, binary and continuous variables were harmonised to maintain consistency across feature sets. This preprocessing ensures the resulting analytical dataset aligns with both theoretical constructs and computational needs for ML/DL models.

Analytical Pipeline

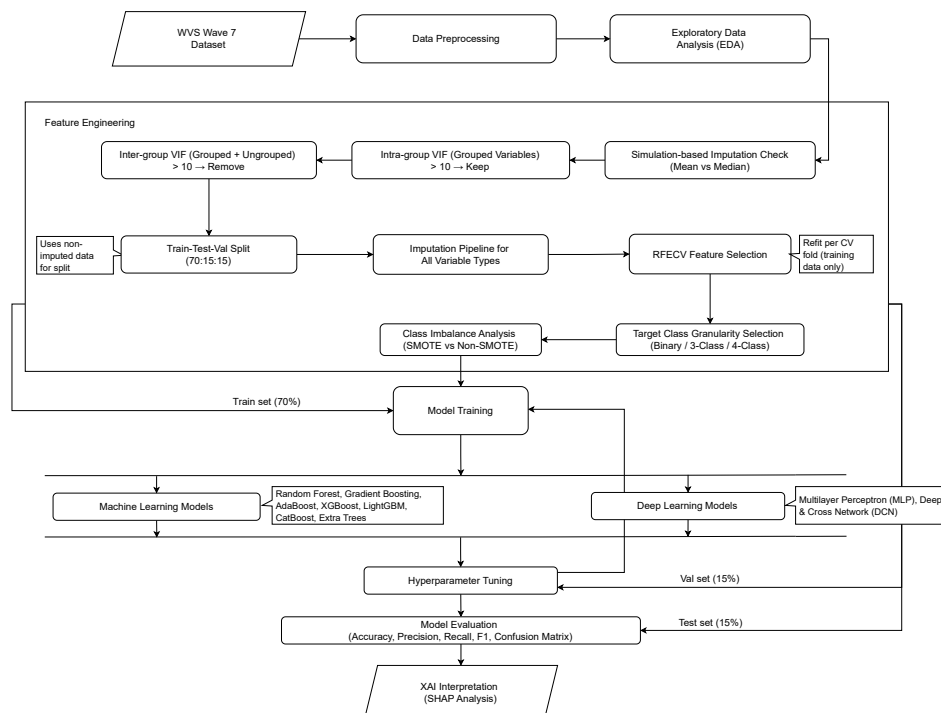


FIGURE 2. *Research Methodology Overview Pipeline (illustrating the end-to-end workflow from data preprocessing to model training, evaluation, and model interpretation).*

The present work applies a rigorous analytic pipeline that seeks to optimise the predictive power at the same time being interpretable (**Figure 2**). The pipeline includes feature engineering, multicollinearity control, data partitioning, model training, evaluation, and interpretability analysis. After exploratory analysis, features were pruned in order to remove redundancy and noise. The multicollinearity was corrected through two-step Variance Inflation factor (VIF): analysis on intra-VIF to determine redundant indicators at each theoretical construct, and analysis on inter-VIF to manage the multicollinearity across constructs. Variables with VIF value above this criterion were not included except where they are retained due to some theoretical reason.

The cleaned dataset was divided into training, validation, and test data sets using a 70:15:15. Missing or invalid entries were imputed while preserving the original feature distributions in order to prevent data leakage and ensure validity for supervised learning. Recursive Feature Elimination using Cross Validation (RFECV) was then used to select the most informative predictors. To determine the impact of outcome granularity, several different target class configurations were considered, such as binary, three-class and four-class happiness classification [30]. Class imbalance was handled by Synthetic Minority Over Sampling Technique (SMOTE). The effect of SMOTE on the predictive performance and stability was assessed with a Random Forest model due to its robustness and reliability when evaluating resampling effects where results with and without the oversampling technique were compared [31].

A comparative analysis was performed using seven ensemble-based machine learning models, i.e., Random Forest, Gradient Boosting, AdaBoost, XGBoost, LightGBM, CatBoost and Extra Trees, and two deep learning architectures, i.e., Multilayer Perceptron (MLP) and Deep & Cross Network (DCN). Hyperparameter tuning was done to optimise each model, where cross-validated grid search (GridSearchCV) was used in ensemble models, and grid search was used in deep learning architectures. Furthermore, the best performed ML and DL models were evaluated by standard classification measures, such as accuracy, precision, recall, F1-score, and confusion matrices. Finally, to connect predictive performance with theoretical understanding, SHapley Additive exPlanations (SHAP) were used on the best-performing model [32]. SHAP values help to quantify the contributions of each feature to the happiness predictions, which can help on building an interpretable Happiness Index based on Maslow's needs and the five dimensions of the Maturity of Heart framework.

To provide a compact and interpretable representation of feature contributions, an aggregate Happiness Index (H_I) was constructed as a weighted linear combination of features, where the weights correspond to normalised mean |SHAP| values.

$$\text{Happiness Index}(H_I) = \sum_{i=1}^n \omega_i x_i, i = 1, \dots, n \quad (1)$$

where:

x_i = standardised feature values

ω_i = normalised mean |SHAP| values for feature i

n = number of selected predictors

In short, given the theoretical foundation and sophisticated data-driven methods, the approach combines both theoretical and practical components to assess the predictive value of motivational and moral-emotional aspects of happiness. The WVS dataset is a globally representative sample whereas the ML/DL pipeline is systematic and interpretable modelling of complex, non-linear relationships. By integrating the Maturity of Heart framework with Maslow's needs, current research is not only able to predict subjective well-being but also measures the contribution each dimension makes to global happiness in different cultures.

RESULTS AND DISCUSSION

The relationships between theoretical constructs chosen and reported happiness were analysed in a preliminary descriptive analysis. The results from Spearman's rank correlation suggested that the variables of core needs developed by Maslow produced the most positive correlations with happiness. Correlation coefficient (r) has values between -1 and $+1$, where values approaching $+1$ represent strong positive relationships, values approaching -1 represent strong negative relationships, and values near zero represent weak or no association. The highest one was the self-actualisation (sact, $r \approx 0.42$) followed by the physiological needs (physio1, $r \approx 0.36$) and then the safety needs (safety1, $r \approx 0.32$). Esteem (esteem5, $r \approx 0.25$) and belonging (belonging1, $r \approx 0.10$) were also positively correlated, but to a lesser extent. Other variables of the Maturity of Heart framework such as civic engagement (civic53, $r \approx 0.19$)

relational trust (trust3, $r \approx 0.13$), ethical reasoning (ethical80, $r \approx 0.13$), personal growth (growth8, $r \approx 0.10$) and empathy/compassion (compassion1, $r \approx 0.08$), had less but clearly positive correlations.

These results confirm the theoretical presupposition that the basic and psychological need satisfaction is one of the fundamental bases of the subjective well-being, and maturity-based socio-moral traits are complementary and context-dependent. Instead of indicating an explicit hierarchy, the findings indicate that various need domain categories and maturity dimensions have an interactive effect on happiness but the extent to which they affect happiness differs according to socioeconomic situations.

RFECV selected 69 optimal predictors, which is a balance between model complexity and predictive accuracy. Class granularity experiments showed that binary classification was the most stable and reliable, with three- and four-class configurations becoming less accurate, presumably because of higher levels of ambiguity and overlap [33], [31]. SMOTE was applied to the training data to achieve higher recall and F1-score of the minority class, reducing class imbalance indicative of its appropriateness to be included in the modelling pipeline [34], [35]. On the one hand, the sensitivity is enhanced through oversampling, however, on the other hand, synthetic patterns can be introduced; hence, the results should be carefully interpreted in connection to minority-class generalisation.

TABLE 1. *Machine Learning and Deep Learning Algorithms' Performance*

Model	Validation Accuracy	Test Accuracy	Validation MCC	Test MCC
CatBoost	0.8793	0.8778	0.4089	0.4029
LightGBM	0.8790	0.8772	0.4046	0.3951
XGBoost	0.8789	0.8780	0.4046	0.4007
Gradient Boosting	0.8787	0.8773	0.4103	0.4032
Random Forest	0.8769	0.8751	0.3704	0.3590
Extra Trees	0.8754	0.8752	0.3462	0.3448
SimpleMLP	0.8667	0.8438	0.3402	0.3508
DCN	0.8425	0.7478	0.2795	0.3032
AdaBoost	0.8035	0.7987	0.4177	0.4112

Seven ensemble-based machine learning models and two deep architecture models were compared in terms of their predictive performance, and the results summarised in **Table 1**. Ensemble models had always performed better than deep learning models both in a validation set and test set. CatBoost had the best validation accuracy (0.8793) and competitive test accuracy (0.8778) and XGBoost had a slightly higher test accuracy (0.8780) but worse validation stability [36]. SimpleMLP and DCN, which are both deep learning models, had worse generalisation, especially DCN with a significant decline between validation and test accuracy.

The Matthews Correlation Coefficient (MCC), which provides a balanced evaluation by considering all elements of the confusion matrix under class imbalance, indicates that ensemble models such as CatBoost achieve consistently strong and stable performance across both validation and test sets. This further supports the selection of CatBoost as the preferred model because it possesses a reliable and balanced predictive capability.

These results are consistent with previous studies demonstrating that tree models often outperform deep learning architectures on structured tabular datasets [37], [38]. This suggests that boosting-based ensemble models are more appropriate to structured survey data with heterogeneous predictors, probably because of their ability to resist non-linear interactions, mixed feature scales, and moderate sample sizes. In contrast, deep learning architectures may require substantially larger datasets or richer feature representations to achieve comparable generalisation performance.

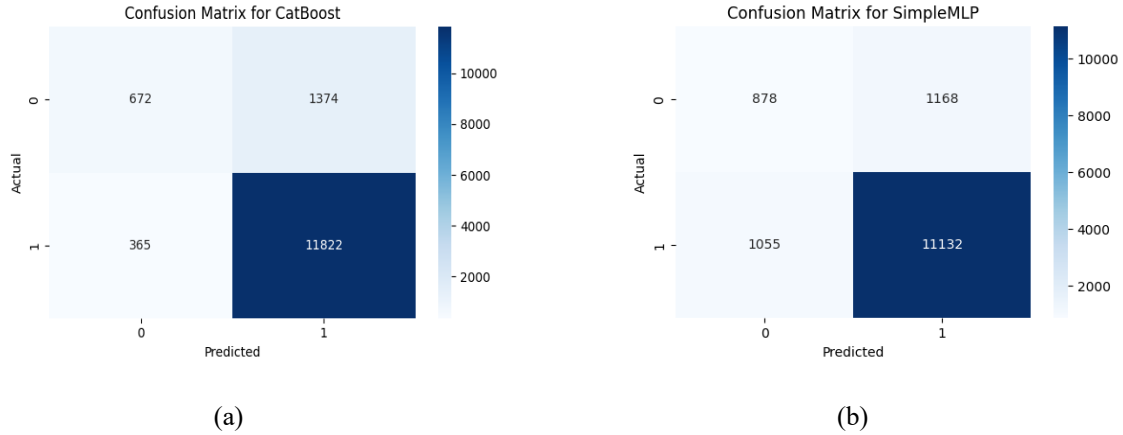


FIGURE 3. *Confusion Matrices for the Best Selected ML and DL Models.*

Figure 3 shows the confusion matrices for the selected ensemble machine learning model (CatBoost) and the deep learning model (SimpleMLP), thus illustrating their predictive performance and error distributions by the classes. Specifically, **Figure 3(a)** shows the confusion matrix for CatBoost, while **Figure 3(b)** presents the results for SimpleMLP. In this study, Class 1 represents respondents with higher subjective well-being (happy), while Class 0 represents respondents with lower levels of happiness.

Overall, CatBoost outperforms SimpleMLP, particularly in identifying positive instances (Class 1). CatBoost correctly classified 11,822 positive cases with a low number of false negatives (365), which means that it has strong ability to detect happiness outcomes. Although it produced more false positives (1,374), this reflects a trade-off favouring higher recall for the positive class.

SimpleMLP in contrast correctly identified 11,132 positive cases but had a higher number of false negatives (1,055), suggesting that it is less sensitive to positive outcomes despite fewer false positives. These results show that CatBoost achieves higher sensitivity toward the positive class and provides a more reliable classification performance for the target outcome.

TABLE 2. *Classification Report Metrics for the Best ML and DL Models*

Model	Class	Precision	Recall	F1-Score	Accuracy
CatBoost	0	0.6480	0.3284	0.4359	0.8778
	1	0.8959	0.9701	0.9305	
	Macro Avg	0.7720	0.6492	0.6837	
	Weighted Avg	0.8602	0.8778	0.8603	
SimpleMLP	0	0.4542	0.4291	0.4413	0.8438
	1	0.9050	0.9134	0.9092	
	Macro Avg	0.6796	0.6713	0.6753	
	Weighted Avg	0.8402	0.8438	0.8420	

Detailed classification metrics further highlighted the superiority of CatBoost. **Table 2** demonstrates that CatBoost attained high precision (0.8959) and recall (0.9701) on positive class (Class 1), thus resulting in a high F1-score of 0.9305. The negative class performance was relatively lower (recall = 0.3284), which is indicative of a modeling bias to be more successful at predicting positive happiness results. Part of this imbalance can be ascribed to the very structure of the classes and the sensitivity prioritisation during the optimisation.

Compared to a naive majority-class baseline of ~85.6%, CatBoost’s 87.78% accuracy represents a meaningful improvement of ~2.2 percentage points, showing that the model captures patterns beyond the class proportions. Macro averages provide an unweighted mean of metrics across both classes, treating each class equally, whereas weighted averages consider the prevalence of the classes, giving more weight to the majority class; CatBoost achieved strong values for both measures, indicating stable performance despite class imbalance. In comparison, SimpleMLP had a lower overall accuracy (0.8438), higher false negatives and lower F1-scores, especially with regard to the positive class. Such findings support the choice of CatBoost as the main model for explainable AI analysis.

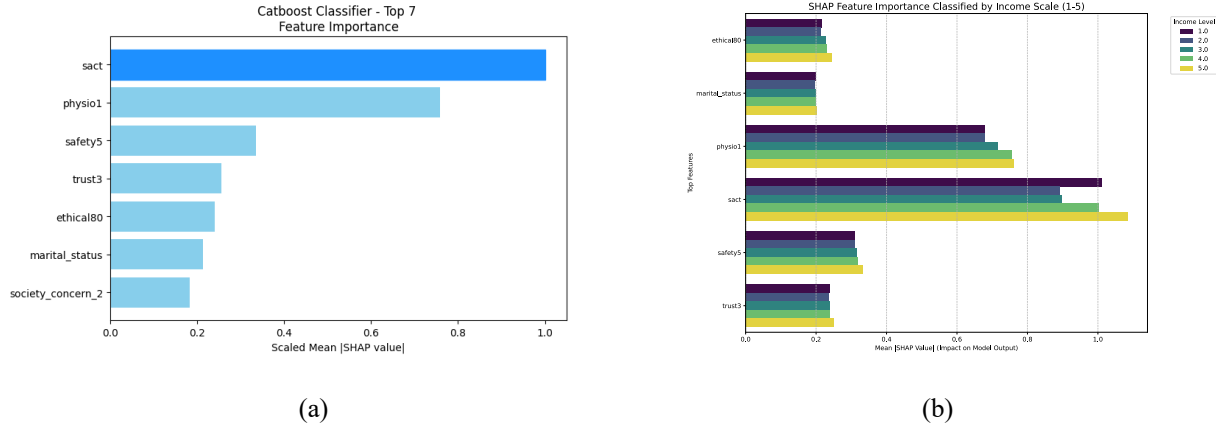


FIGURE 4. SHAP-Based Explainable AI Analysis of Happiness Predictions.

The SHapley Additive exPlanations (SHAP) were calculated to understand the predictions of CatBoost, and features were ranked by scaled mean |SHAP| values (max=1) to determine how strongly they contributed to model predictions. The emergence of self-actualisation (sact) as the most powerful predictor was as demonstrated in **Figure 4(a)** followed by physical well-being (physio1) and perceived safety (safety5). Maturity of Heart related features, like relational trust (trust3) and ethical orientation (ethical80), also played a role, but to a smaller degree, and helped to refine the predictions without dominating them. This trend is consistent with the theoretical expectations which support primacy of core needs and modulatory function of moral and relational maturity.

To further enhance the interpretability, these five most influential predictors were combined into a compact Happiness Index (H_I), which aggregates standardised feature values weighted by their normalised mean |SHAP| contributions. Higher H_I values correspond to greater predicted levels of subjective well-being, which provide an intuitive summary of the joint effects of the main psychological and maturity-related factors on the outcomes of happiness. This index is not a standalone predictive model but offers a concise, interpretable representation that aligns with both theoretical constructs and model insights.

On top of the global importance of features, SHAP values provide a more detailed insight into the impact of the values of individual features on model predictions [39]. SHAP not only shows the direction; it also shows the level of contribution of each feature, therefore, showing whether each variable increases or decreases the likelihood of being assigned into the higher happiness group or not. As an example, a high level of self-actualisation and perceived safety have a consistent positive contribution to predictions of happiness, and low levels have a negative impact on the classification of happiness. Therefore, the model not only determine salient predictors but also explains how the differences in psychological and maturity-related factors influence the outcomes of individual prediction. Even though the existing visualisation highlights the importance of aggregated features, the directional effects are obtained based on the SHAP values calculated on the observation level.

A stratified SHAP analysis based on income also showed that there were systematic variations in the importance of features between socioeconomic groups as shown in **Figure 4(b)**. Among the leading predictors, the largest SHAP magnitudes were those of the highest income group (Class 5), meaning that it was more sensitive to changes in the need satisfaction and maturity-related features. Self-actualisation and physical well-being had significant effects on the impact, which rise with income, while marital status remained consistently significant across income levels. Perceived safety was found to be a salient predictor in all income groups with relatively higher SHAP values in the highest income group. It is important to note that the effects of trust and ethical orientation were stronger among those who have higher income levels. This pattern indicates that, when material security is enhanced, an increase in socio-moral and relational characteristics is more likely to explain happiness, but does not imply enhanced preferences or values. Comprehensively, the stratified analysis illustrates the ability of the model to differentiate universal determinants of happiness and socioeconomically contingent effects.

In spite of these advantages, there are a few limitations that must be considered. The paper is based on cross-sectional survey data, which limits causality interpretation [6]. There is also the possibility of self-reported measures creating bias in answers, and the indicators chosen, though theoretically based, might be inadequate to represent the complexity of the psychological and moral constructs [40]. Furthermore, the focus on classification accuracy may

underrepresent nuanced subjective well-being gradients, as it reduces happiness into discrete categories and may overlook subtle variations in individual well-being levels. These limitations highlight important directions for extending the present findings, particularly in capturing temporal dynamics and finer-grained well-being variation.

In general, the results indicate that boosting-based ensemble models, specifically CatBoost, give strong and generalisable predictions of happiness [32]. The explainable AI techniques provide interpretable results that can mediate between performances of machine learning and the established psychological theory, and the objective interpretation of subjective well-being in terms of both need satisfaction and maturity-related aspects.

CONCLUSION

This study proves that by combining ensemble and deep learning models to cutting-edge psychological theories, one can obtain a solid predictive performance and at the same time, meaningful theoretical understanding of subjective happiness. Besides the accuracy of classification, the main contribution it makes is the creation of a machine learning pipeline that is interpretable and able to model complex psychosocial data in a survey and maintain theoretical coherence. Through the confluence of Maslow's needs with the Maturity of Heart model and the SHapley Additive exPlanations (SHAP), the suggested solution will fill the gap between data-driven prediction and human motivation theory, allowing building a transparent and theory-based Happiness Index. As shown, the findings do not point to the fact that only the satisfaction of basic or psychological needs can define happiness, but also point to the moral, emotional and relational maturity dimension. Such a combined view makes machine learning models more explanatory and allows their use in cross-cultural settings, educational settings and policy settings, where interpretability and theoretical consistency are needed.

Future Study: Future research needs to build on the current study by using longitudinal designs to help in the measurement of dynamic changes in determinant of happiness with time. The proposed Happiness Index may be further refined by alternative operationalisations of maturity related constructs and outcome formulations that are multi-hybrid, multi-regression based or a combination of the two. Moreover, the idea of using culturally calibrated weighting schemes and more balance datasets can enhance the generalisability to different populations [31]. Lastly, by making a step beyond feature attribution and into the social science theory-based methods of explanation, future models would be able to offer contrastive and interaction-based information and thus become more practically useful in terms of psychological testing and intervention.

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